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CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2009 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 2

*Time allowed - 3 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided.
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 8.

*****Each page must show your name and your class. *****

QUESTION ONE.

(15 MARKS)

MARKS

- (a) Evaluate $\int_0^3 \frac{x}{\sqrt{16+x^2}} dx$. 3
- (b) Find $\int \frac{x^2+x+1}{x(x^2+1)} dx$. 2
- (c) By using the substitution $t = \tan \frac{\theta}{2}$, evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{2+\sin\theta}$. 3
- (d) Find $\int x e^{-x} dx$. 2
- (e) (i) If $I_n = \int_0^1 (1+x^2)^n dx$, $n = 0, 1, 2, \dots$ show that 3
 $(2n+1) I_n = 2^n + 2n I_{n-1}$ for $n = 1, 2, \dots$
- (ii) Hence find a reduction formula for $J_m = \int_0^{\frac{\pi}{4}} \sec^{2m} x dx$. 2

QUESTION TWO.

(15 MARKS)

(START A NEW PAGE)

- (a) If z is such that $|z| = 3$ and $\arg z = \frac{\pi}{3}$, mark on the same Argand diagram 4
 (i) z (ii) $\bar{z}$ (iii) iz (iv) z^{-1}
- (b) In an Argand Diagram, the point P representing the complex number z moves 1
 so that $|z - (1+i)| = 1$.
- (i) Sketch the locus of P 1
- (ii) Shade the region where $|z - (1+i)| = 1$ and $0 < \arg(z-1) < \frac{\pi}{4}$. 1

Question 2 continued on next page.

QUESTION TWO CONTINUED.

MARKS

(c) Given $z = \cos\theta + i\sin\theta$ and for the positive integers n ,

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i\sin n\theta.$$

(i) Expand $(z + \frac{1}{z})^4 + (z - \frac{1}{z})^4$ to show that $\cos^4\theta + \sin^4\theta = \frac{1}{4}(\cos 4\theta + 3)$ 2

(ii) By letting $x = \cos\theta$, show that the equation $8x^4 + 8(1 - x^2)^2 = 7$ 2

$$\text{has roots } \pm \cos \frac{\pi}{12}, \quad \pm \cos \frac{5\pi}{12}$$

(iii) Show that $\cos \frac{\pi}{12} \cdot \cos \frac{5\pi}{12} = \frac{1}{4}$ and 3

$$\cos \frac{\pi}{12} + \cos \frac{5\pi}{12} = \sqrt{\frac{3}{2}}$$

(iv) Hence or otherwise, find a surd expression for $\cos \frac{\pi}{12}$ 2

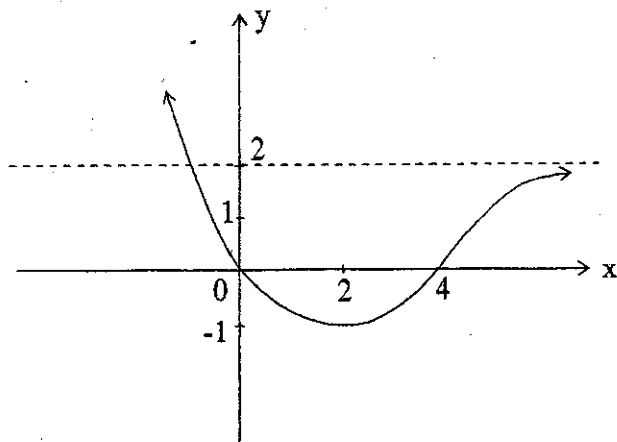
QUESTION THREE.

(15 MARKS)

(START A NEW PAGE)

MARKS

- (a) The diagram shows the graph of $f(x)$.



Sketch on separate diagrams, the following curves, indicating clearly any turning points and asymptotes.

(i) $y = \left| \frac{1}{f(x)} \right|$

2

(ii) $y = (f(x))^2$

2

(iii) $y^2 = f(x)$

2

(iv) $y = xf(x)$

2

- (b) (i) Show that the area enclosed by a parabola $x^2 = 4ay$

3

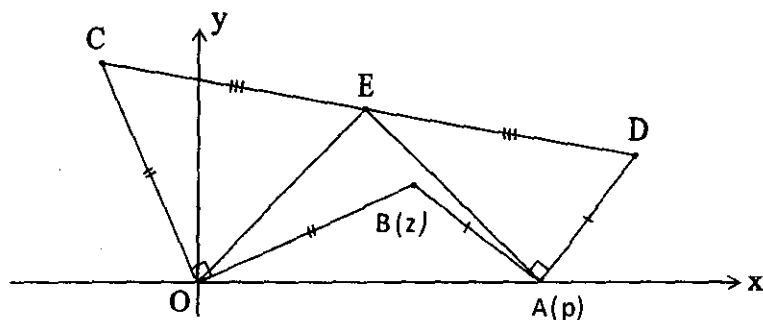
and its latus rectum is given by $A = \frac{8a^2}{3} \text{ units}^2$.

- (ii) A solid is formed such that its base is a semicircle of radius one metre. Vertical sections parallel to the diameter are parabolas with each latus rectum being a chord of the semicircle parallel to the diameter. By using the result from (i) and the technique of slicing, find the volume of this solid.

4

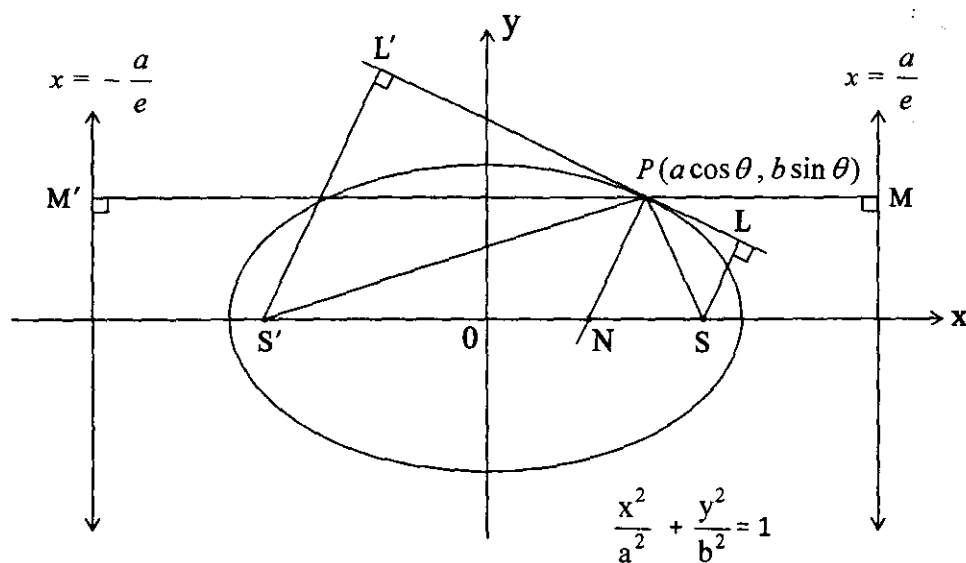
- (a) In the Argand diagram O, A and B represent the origin, the number p and the complex number z respectively. By rotating B about A by 90° in a clockwise direction we get the point D and by rotating B about O in an anticlockwise direction we get the point C. Let E be the midpoint of CD.

3



Show that $\triangle OEA$ is a right angled isosceles triangle with a right angle at E.

- (b) Lines drawn from the foci S and S' of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, are perpendicular to the tangent drawn at $P(a \cos \theta, b \sin \theta)$. They meet this tangent respectively at L and L' . The line parallel to the x -axis passing through P intersects the directrices at M and M' and the normal at P meets the x -axis at N .



Question 4 continued on next page.

QUESTION 4(b) CONTINUED.

MARKS

- (b) (i) Show that $PS = a(1 - e \cos \theta)$ 2
- (ii) Write down a similar expression for PS' . 1
- (iii) Show that the equation of the tangent at P is 2
- $$bx \cos \theta + ay \sin \theta - ab = 0$$
- (iv) Find the distances SL and SL' from the foci S and S' to the tangent at P . 2
- (v) Hence, or otherwise, show that PN bisects $\angle SPS'$. 3
- (vi) Show that $\frac{PS}{NS} = \frac{PS'}{NS'}$ 2

QUESTION FIVE.

(15 MARKS)

(START A NEW PAGE)

- (a) Given that a, b, c and d represent positive integers and that $a + b + c = 3d$ show that $100a + 10b + c$ is divisible by 3. 2
- (b) The roots of $x^3 + 3px + q = 0$ are α, β and γ , (none of which are equal to 0).
- (i) Find the monic equation with roots $\frac{\beta\gamma}{\alpha}, \frac{\alpha\gamma}{\beta}$ and $\frac{\alpha\beta}{\gamma}$, giving the coefficients in terms of p and q . 4
- (ii) Deduce that if $\gamma = \alpha\beta$ then $(3p - q)^2 + q = 0$. 2
- (c) Determine the values of a and b given that $(x + 1)^2$ is a factor of $P(x) = x^5 + 2x^2 + ax + b$. 3
- (d) Show that $\frac{\sin(2k+1)\alpha}{\sin \alpha} - \frac{\sin(2k-1)\alpha}{\sin \alpha} = 2\cos(2k\alpha)$ if $0 < \alpha < \frac{\pi}{2}$. 2
- (e) Given that $\sin^{-1}x, \cos^{-1}x$ and $\sin^{-1}(1-x)$ are acute, show that 2
- $$\sin(\sin^{-1}x - \cos^{-1}x) = 2x^2 - 1.$$

- (a) A toy car of mass 1 kg , initially at rest, starts to move by a propelling force of 50 N , provided by its engine along a straight road.

The car experiences a resistance force of kv^2 Newtons, where v is its velocity in metres per second and k is a positive constant.

The limiting velocity of the car is 10 m/s .

Let x be the displacement of the car at time t seconds after it starts to move.

(i) Show that $2 \frac{d^2x}{dt^2} = 100 - v^2$ 2

(ii) Show that $v^2 = 100(1 - e^{-x})$ 2

(iii) Find the time taken for the car to reach a velocity of 5 m/s . 3

(iv) When the car reaches a velocity of 5 m/s the engine is switched off and a braking force of F is applied. Find F , given that the distance travelled by the car to stop is equal to the distance to reach 5 m/s . 3

- (b) The depth of water in a harbour on a particular day is 8.2 metres at low tide and 14.6 metres at high tide. Low tide is at $1:05\text{ pm}$ and high tide is at $7:20\text{ pm}$. The captain of a ship wants to leave the harbour after midday on that day. To leave the harbour the ship requires at least 13.3 metres of water. 5

Find between what two times of that day the captain can leave the harbour.

- (a) The part of the curve $y = \ln \frac{x}{e}$ between $x = e$ and $x = e^2$ is rotated about the x -axis to form a solid.
- (i) Draw the curve $y = \ln \frac{x}{e}$ between $x = e$ and $x = e^2$ and show a sketch of the solid formed by the rotation of this curve. 1
- (ii) Use the method of cylindrical shells to find the volume of the solid. 4
- (b) A body is projected vertically upwards from the surface of the Earth with initial speed u . The acceleration due to gravity, g at any point on its path is inversely proportional to the square of the distance from the centre of the Earth. R is the radius of the Earth.
- (i) Prove that the speed v at any position x is given by 3
- $$v^2 = u^2 + 2gR^2\left(\frac{1}{x} - \frac{1}{R}\right)$$
- (ii) Prove that the greatest height H above the Earth's surface is 2
- given by $H = \frac{u^2 R}{2gR - u^2}$
- (iii) Show that the body will escape from the Earth if $u \geq \sqrt{2gR}$ 1
- (iv) Find the minimum speed in km/s with which the body must be initially projected from the surface of the Earth so as to never return. 1
- (Take $R = 6400km, g = 10m/s^2$)
- (v) If $u = \sqrt{2gR}$ prove that the time taken to reach a height $3R$, 3
- above the surface of the Earth is equal to $\frac{14}{3} \sqrt{\frac{R}{2g}}$.

QUESTION EIGHT.

(15 MARKS)

(START A NEW PAGE)

MARKS

(a) (i) Sketch the graph of $y = px^2 + q$ where p and q are positive constants.

1

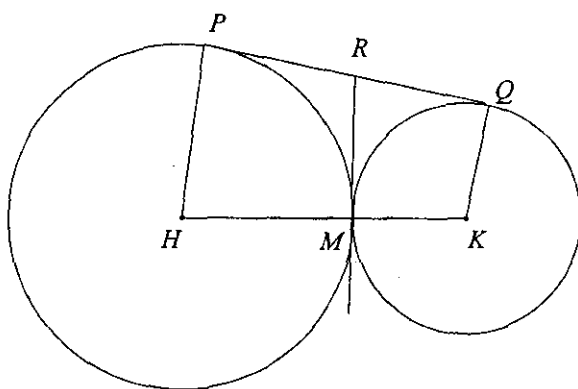
(ii) By considering the area represented by $\int_1^2 (px^2 + q)dx$, show that

2

$$p + q < \frac{7p+q}{3} < 4p + q$$

(b) Shown are two circles centres H and K which touch at M .

PQ and RM are common tangents.



(i) Show that quadrilaterals $HPRM$ and $MRQK$ are cyclic.

2

(ii) Prove that triangles PRM and MKQ are similar.

3

Question 8 continued on the next page.

QUESTION EIGHT CONTINUED.

- (c) The Taylor series provides a way of expressing certain functions as infinite series.
Using the Taylor series, it can be shown that

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$e^{ix} = 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \dots$$

- (i) Using these infinite series, show that $e^{i\pi} = -1$ 2
- (ii) Show i^i is a real number. 2
- (iii) Find the general formula for $\ln z$ where $z = x + iy$, and state the interval for $\ln z$ that will give the principle value of $\ln z$. 2
- (iv) Hence, find the principle value for the complex number $z = 1 + i$ 1

END OF TEST.

Solutions:

(1)

2009

Question One: (15 Marks)

(a) Let $u = 16 + x^2$

$$\frac{du}{dx} = 2x$$

$$\therefore du = 2x dx$$

$$\text{IF } \begin{cases} x=0 & u=16 \\ x=3 & u=25 \end{cases}$$

(1)

$$\int_0^3 \frac{x}{\sqrt{16+x^2}} dx = \int_{16}^{25} \frac{\frac{1}{2} du}{\sqrt{u}}$$

ALTERNATIVE:

OR $x = 4 \tan \theta$, $dx = 4 \sec^2 \theta d\theta$

$$\int_0^{\tan^{-1} \frac{3}{4}} \frac{4 \tan \theta}{\sqrt{16+16 \tan^2 \theta}} \cdot 4 \sec^2 \theta d\theta$$

$$= \int_0^{\tan^{-1} \frac{3}{4}} 4 \tan \theta \sec \theta d\theta$$

$$= 4 \left[\sec \theta \right]_0^{\tan^{-1} \frac{3}{4}}$$

$$= 4 \left[\sec \left(\tan^{-1} \left(\frac{3}{4} \right) \right) - \sec 0 \right]$$

$$= 4 \left[\frac{5}{4} - 1 \right] = 1$$

$$= \frac{1}{2} \int_{16}^{25} u^{-1/2} du$$

$$= \frac{1}{2} \left[2u^{1/2} \right]_{16}^{25}$$

$$= \left[\sqrt{u} \right]_{16}^{25}$$

$$= \sqrt{25} - \sqrt{16}$$

$$= 1$$

$$b) \int \frac{x^2 + x + 1}{x(x^2 + 1)} dx = \int \frac{(x^2 + 1) + x}{x(x^2 + 1)} dx = \int \left(\frac{1}{x} + \frac{1}{x^2 + 1} \right) dx$$

$$= \ln|x| + \tan^{-1} x + C$$

$$c) \begin{cases} t = \tan \theta/2 \\ \frac{dt}{d\theta} = \frac{1}{2} \sec^2 \theta/2 \\ dt = \frac{1}{2} (1+t^2) d\theta \\ \text{Hence } d\theta = \frac{2}{(1+t^2)} dt \end{cases}$$

$$\begin{cases} \text{When } \theta=0 & t=0 \\ \theta=\pi & t=1 \end{cases}$$

$$\text{Also } \sin \theta = \frac{2t}{1+t^2}$$

$$\int_0^{\pi/2} \frac{d\theta}{2 + \sin \theta} =$$

$$= \int_0^1 \frac{2}{1+t^2} \times \frac{1}{2 + \frac{2t}{1+t^2}} dt$$

(2)

$$= \int_0^1 \frac{dt}{t^2 + t + 1}$$

$$= \int_0^1 \frac{dt}{(t + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right] = \frac{2}{\sqrt{3}} \left[\frac{\pi}{3} - \frac{\pi}{6} \right] = \frac{\pi}{3\sqrt{3}}$$

$$\text{OR } = \frac{\sqrt{3}\pi}{9} \div 0.6045$$

$$d) \int x e^{-x} dx = \int x \frac{d}{dx} (-e^{-x}) dx$$

$$= -x e^{-x} - \int 1 \cdot (-e^{-x}) dx$$

$$= -x e^{-x} + \int e^{-x} dx$$

$$= -x e^{-x} - e^{-x} + C$$

$$= e^{-x} (-x - 1) + C \text{ or } -e^{-x} (x + 1) + C$$

$$e) I_n = \int_0^1 (1+x^2)^n dx$$

$$\textcircled{1} \rightarrow = \left[x(1+x^2)^n \right]_0^1 - \int_0^1 x \cdot n(1+x^2)^{n-1} \cdot 2x dx$$

$$= 2^n - 2n \int_0^1 x^2 (1+x^2)^{n-1} dx$$

$$\textcircled{1} \rightarrow = 2^n - 2n \int_0^1 (1+x^2-1)(1+x^2)^{n-1} dx$$

$$= 2^n - 2n \left\{ \int_0^1 (1+x^2)^n dx - \int_0^1 (1+x^2)^{n-1} dx \right\}$$

$$\textcircled{1} \rightarrow I_n = 2^n - 2n I_n + 2n I_{n-1}$$

$$\therefore (2n+1) I_n = 2^n + 2n I_{n-1}, n=1, 2, 3$$

(ii) $u = \tan x$ $x=0 \quad u=0$
 $du = \sec^2 x dx$ $x=\pi/4 \quad u=1$

$$I_m = \int_0^{\pi/4} \sec^{2m} x dx$$

$$= \int_0^{\pi/4} (\sec^2 x)^{m-1} \sec^2 x dx$$

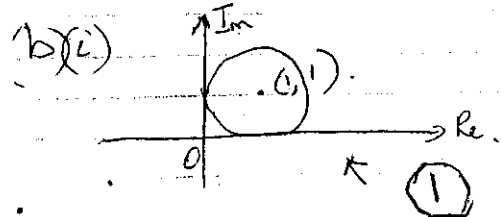
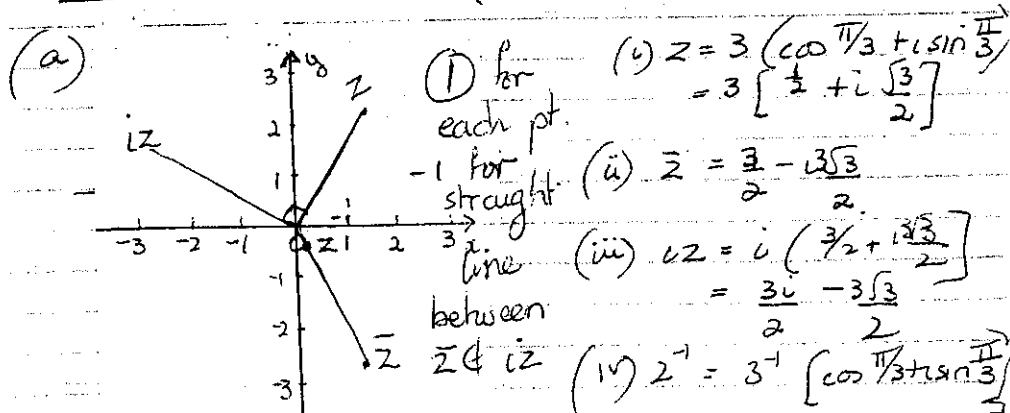
① $\rightarrow = \int_0^1 (1+u^2)^{m-1} du$

$\therefore I_m = I_{m-1} \quad m=1, 2, 3, \dots$

$$\{2(m-1)+1\} I_m = 2^{m-1} + 2(m-1) I_{m-2}$$

① $\rightarrow \therefore (2m-1) I_m = 2^{m-1} + 2(m-1) I_{m-2}$
 $m=2, 3, 4, \dots$

Question Two: (15 Marks)



(c)(i) $\left(z + \frac{1}{z} \right)^4 + \left(z - \frac{1}{z} \right)^4 = 2 \left(z^4 + 6z^2 \cdot \frac{1}{z^2} + \frac{1}{z^4} \right)$
 $= 2 \left(z^4 + \frac{1}{z^4} \right) + 12$

① $\rightarrow (2 \cos \theta)^4 + (2 \sin \theta)^4 = 2(2 \cos 4\theta) + 12$
 $16 (\cos^4 \theta + \sin^4 \theta) = 4(\cos 4\theta + 3)$
 $\therefore \cos^4 \theta + \sin^4 \theta = \frac{1}{4} (\cos 4\theta + 3)$

(ii) $x = \cos \theta$
 $8x^4 + 8(1-x^2)^2 = 7$
 $1-x^2 = \sin^2 \theta$
 $8(\cos^4 \theta + \sin^4 \theta) = 7$
 $2(\cos 4\theta + 3) = 7$
Hence eqn. becomes
 $x = \cos \theta, \cos 4\theta = \frac{1}{2}$ ①
 $4\theta = 2n\pi \pm \frac{\pi}{3}$
 $\theta = \frac{(6n \pm 1)\pi}{12} \quad n=0, \pm 1, \pm 2, \dots$

$x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}, \cos \frac{11\pi}{12}$
 $x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos(\pi - \frac{5\pi}{12}), \cos(\pi - \frac{\pi}{12})$
 $\therefore x = \pm \cos \frac{\pi}{12}, \pm \cos \frac{5\pi}{12}$

> 5-

(iii) $8x^4 + 8(1-x^2)^2 = 7$

$16x^4 - 16x^2 + 1 = 0$

roots: $\cos \frac{\pi}{12}, -\cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, -\cos \frac{5\pi}{12}$

① then $\alpha\beta\gamma\delta = \cos^2 \frac{\pi}{12} \cos^2 \frac{5\pi}{12} = \frac{1}{16}$
 $\sum \alpha\beta = -\cos^2 \frac{\pi}{12} - \cos^2 \frac{5\pi}{12} = -\frac{1}{4}$

① where $0 < \frac{\pi}{12} < \frac{5\pi}{12} < \frac{\pi}{2}$

→ Then $\cos \frac{\pi}{12} \cos \frac{5\pi}{12} = +\sqrt{\frac{1}{16}} = \frac{1}{4} \neq$

① → $\cos^2 \frac{\pi}{12} + \cos^2 \frac{5\pi}{12} + 2\cos \frac{\pi}{12} \cos \frac{5\pi}{12} = 1 + \frac{1}{2}$
 $\therefore (\cos \frac{\pi}{12} + \cos \frac{5\pi}{12})^2 = \frac{3}{2}$
 $\cos \frac{\pi}{12} + \cos \frac{5\pi}{12} = \frac{\sqrt{3}}{2}$

(iv) $\cos \frac{\pi}{12}, \cos \frac{5\pi}{12}$ are roots of the quad. eqn.
 $x^2 - \frac{\sqrt{3}}{2}x + \frac{1}{4} = 0$

① → $x = \frac{\frac{\sqrt{3}}{2} \pm \sqrt{\frac{3}{4} - 1}}{2} = \frac{\sqrt{3} \pm 1}{2\sqrt{2}}$ ①

$\cos \frac{\pi}{12} > \cos \frac{5\pi}{12} \Rightarrow \cos \frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}}$

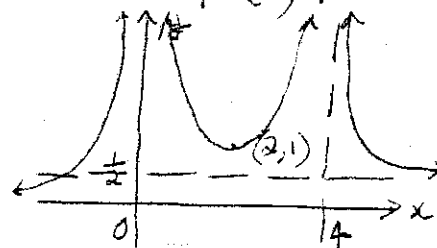
OR 0.9659

OR $\frac{\sqrt{6}+\sqrt{2}}{4}$

QUESTION THREE:

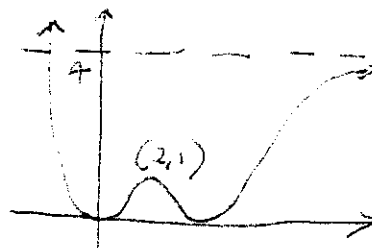
-6-

(a) (i) $y = \left| \frac{1}{f(x)} \right|$



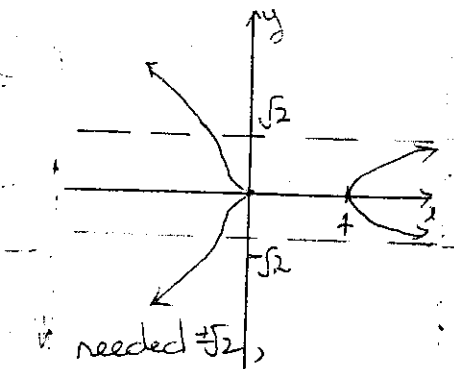
- ① for 3 branches
- ① for asymptotes & turning pt.

(ii) $y = [f(x)]^2$



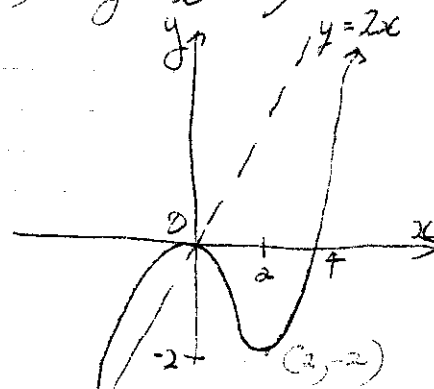
- ① for shape
- ① for T.P & y = 4 asympt.

(iii) $y^2 = f(x)$

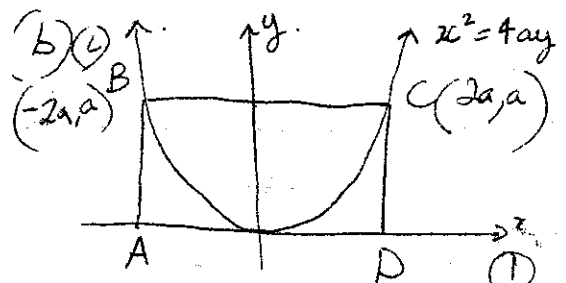


needed ± 52

(iv) $y = x f(x)$



- ① for shape
- ① for asympt y = 2x.



length of LR = 4a
 Area is area of rectangle ABCD - area under the curve between $x = -2a$ and $x = 2a$

① - 7 -

$$\begin{aligned} \text{Area} &= 4a \cdot a - \int_{-2a}^{2a} \frac{x^2}{4a} dx \quad \leftarrow \text{①} \\ &= 4a^2 - 2 \left[\frac{x^3}{12a} \right]_{-2a}^{2a} \\ &= 4a^2 - 2 \left[\frac{8a^3 - 0}{12a} \right] = 4a^2 - \frac{4a^2}{3} \end{aligned}$$

* Could do this question next to the y axis eg $\frac{2}{3}ay^2$. $\therefore A = \frac{8a^2}{3}u^2$

(ii) parabola

Let slice be Δk thick
When $y = k$
 $x = \pm \sqrt{1-k^2}$
 $\therefore$ length of L-R = $2\sqrt{1-k^2}$
So $4a = 2\sqrt{1-k^2}$
 $a = \frac{\sqrt{1-k^2}}{2} \quad \leftarrow \text{①}$

Using (i) Area of section = $\frac{8}{3} \left(\frac{\sqrt{1-k^2}}{2} \right)^2$
 $= \frac{8(1-k^2)}{12}$
 $= \frac{2(1-k^2)}{3} \quad \leftarrow \text{①}$

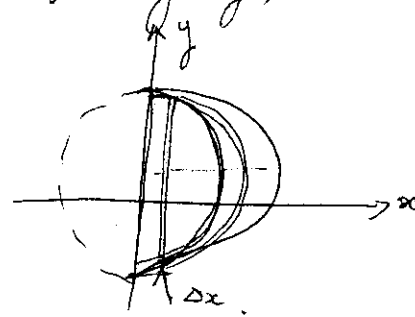
Volume of the slice = $\frac{2(1-k^2)}{3} \Delta k$

Vol. of Solid = $\frac{2}{3} \int_0^1 (1-k^2) dk$
 $= \frac{2}{3} \left[k - \frac{k^3}{3} \right]_0^1 \quad \leftarrow \text{①}$
 $= \frac{2}{3} \left[1 - \frac{1}{3} \right] = \frac{4}{9}$
 $= \frac{4}{9} m^3 \quad \leftarrow \text{①}$

Alternate solution for (Q3b ii)

(7a)

At height y , then thickness is Δx



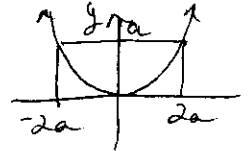
$$\text{Area} = \frac{8}{3} a^2$$

"a" is half the y-distance

$$= \left(\frac{y}{2} \right)^2 \times \frac{8}{3}$$

$$= \frac{y^2}{4} \times \frac{8}{3}$$

$$= \frac{2}{3} y^2 \quad \leftarrow \text{①}$$



① $\rightarrow$ So $dV = \frac{2}{3} y^2 \Delta x$ (Area of parabola $\times$ thickness)

$$V = \int_0^1 \frac{2}{3} \cdot (1-x^2) dx$$

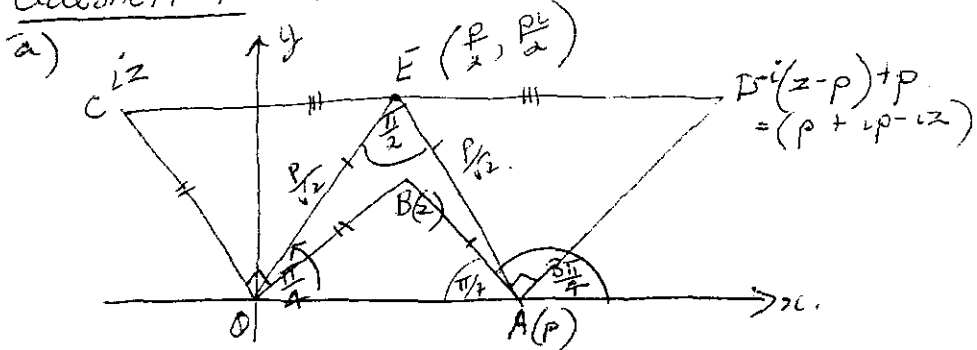
$$= \frac{2}{3} \left[x - \frac{x^3}{3} \right]_0^1 \quad \leftarrow \text{①}$$

$$= \frac{2}{3} \times \left[1 - \frac{1}{3} \right]$$

$$= \frac{4}{9}$$

①

Question 4 Alternate Solution (8b)



$\vec{AB} = z - p$
 $\vec{AD}$ is the pt $-i(z-p)$ So $\vec{OD} = \vec{OA} + \vec{AD}$
 $= p + [-i(z-p)]$
 $= -iz + ip + p$

Since E is the midpt of CD it has co-ords.
 $\frac{iz - i(z-p) + p}{2} = \frac{p}{2} + \frac{p}{2}i$

Distance of OE:
 $\sqrt{\left(\frac{p}{2}\right)^2 + \left(\frac{p}{2}\right)^2}$
 $= \sqrt{\frac{p^2}{4} + \frac{p^2}{4}}$
 $= \sqrt{\frac{p^2}{2}} = \frac{p}{\sqrt{2}}$

Distance of EA:
 $\sqrt{\left(p - \frac{p}{2}\right)^2 + \left(\frac{p}{2}\right)^2}$
 $= \sqrt{\frac{p^2}{4} + \frac{p^2}{4}}$
 $= \sqrt{\frac{p^2}{2}} = \frac{p}{\sqrt{2}}$

$\therefore OE = EA$
 $\therefore \triangle OEA$ is
 isos $\triangle$.

The arg(E) = $\tan^{-1}\left(\frac{p/2}{p/2}\right)$
 $= \tan^{-1}(1) = \frac{\pi}{4}$
 $\therefore \angle EAO = \frac{\pi}{4}$ (is isos $\triangle$)
 $\therefore \angle OEA = 90^\circ = \frac{\pi}{2}$ (sum of a triangle)

QUESTION FOUR (8)

(a) $\vec{AB}$ represents $z - p$
 $\vec{AD}$ is the clockwise rotation of $\vec{AB}$ by 90°
 $\therefore \vec{AD}$ represents $-i(z-p)$
 $\vec{OD} = \vec{OA} + \vec{AD} = p + [-i(z-p)]$
 $= -iz + ip + p$
 $= p + z + i(p-z)$

Let $z = x + iy$ $\therefore L$ is the pt $(p + y, p - x)$
 $\vec{OE}$ is the anticlockwise rotation of $\vec{OB}$ by 90°
 and $\vec{OB}$ represents $iz = i(x + iy) = -y + ix$
 $\therefore E$ is the pt $(-y, x)$

Since E is the midpt of CD it has co-ord. $\left(\frac{p}{2}, \frac{p}{2}\right)$
 Let R be the midpt of OA, directly below E. $\therefore E$ is the $\perp$ bisector of OA the $\triangle ORE$ is isos. and as a circle centred at R of radius $\frac{p}{2}$ can pass through the pts O, E & A then $\angle OAE$ is 90° at E.

b) (i) $\frac{PS}{PM} = e \therefore PS = e PM = e \left(\frac{a}{e} - a \cos \theta\right)$
 $= a(1 - e \cos \theta)$

(ii) Similarly $\frac{PS'}{PM'} = e \therefore PS' = e PM' = e \left(\frac{a}{e} + a \cos \theta\right)$
 $= a(1 + e \cos \theta)$

(iii) Differentiating $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 $\frac{2x}{a^2} + \frac{2y}{b^2} \times y' = 0$
 $y' = -\frac{xb^2}{ya^2} = -\frac{b \cos \theta}{a \sin \theta}$

Equation of tangent at P
 $y = b \sin \theta = \frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

(i) $ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$
 $bx \cos \theta + ay \sin \theta - ab(\sin^2 \theta + \cos^2 \theta) = 0$
 $bx \cos \theta + ay \sin \theta - ab = 0$

9.

(iv) Distance from $a(ae, 0)$ to tangent is
 $SL = \frac{|abe \cos \theta - ab|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} = \frac{ab(e \cos \theta - 1)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$

$$\frac{ab(1 - e \cos \theta)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \quad \text{--- (1)}$$

Analogously $SL' = \frac{|-abe \cos \theta - ab|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} = \frac{ab(1 + e \cos \theta)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \quad \text{--- (1)}$

(v) In $\triangle PS'L'$ & $\triangle PSL$
 $\frac{PS'}{PS} = \frac{a(1 + e \cos \theta)}{a(1 - e \cos \theta)} = \frac{1 + e \cos \theta}{1 - e \cos \theta}$

$$\frac{SL'}{SL} = \frac{1 + e \cos \theta}{1 - e \cos \theta}$$

(1) $\therefore \frac{PS'}{PS} = \frac{SL'}{SL} \quad \text{--- [eqn 1]}$

$\sin \angle L'PS' = \frac{SL'}{PS'} = \frac{SL}{PS} \quad \text{from [eqn]}$

(1) $\therefore \angle L'PS' = \angle LPS$

$\therefore \angle S'PN = 90^\circ - \angle L'PS' = 90^\circ - \angle LPS = \angle SPN$

(1) $\therefore PN$ bisects $\angle SPS'$

(vi) PN has gradient $= \frac{a \sin \theta}{b \cos \theta}$
 Eqn of PN is $y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$

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x -intercept when $y=0$

(1) $x = \frac{(a^2 - b^2) \sin \theta \cos \theta}{a \sin \theta} = \frac{(a^2 - b^2)(1 - e^2) \cos \theta}{2a}$

$\rightarrow = ae^2 \cos \theta$
 $NS = ae - ON = ae - ae^2 \cos \theta = ae(1 - e \cos \theta)$

$NS' = ae + ae^2 \cos \theta = ae(1 + e \cos \theta)$

$\therefore \frac{NS}{NS'} = \frac{1 - e \cos \theta}{1 + e \cos \theta} = \frac{PS}{PS'}$

(1) $\therefore \frac{PS}{NS} = \frac{PS'}{NS'} \quad \text{(by (i) & (ii) parts)}$

QUESTION : FIVE

(a) $100a + 10b + c = 99a + 9b + a + b + c \quad \text{--- (1)}$

$= 9(11a + b) + 3d$

$= 3(3(11a + b) + d) \quad \text{--- (1)}$

Hence divisible by 3.

(b) (1) $\alpha + \beta + \gamma = 0, \alpha\beta + \alpha\gamma + \beta\gamma = 3p, \alpha\beta\gamma = -q$

$\therefore \frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} + \frac{\alpha\beta}{\gamma} = \frac{(\beta\gamma)^2 + (\alpha\gamma)^2 + (\alpha\beta)^2}{\alpha\beta\gamma}$

(1) $= \frac{(\beta\gamma + \alpha\gamma + \alpha\beta)^2 - 2(\alpha\beta\gamma^2 + \alpha\gamma\beta^2 + \alpha\beta\gamma^2)}{\alpha\beta\gamma}$

$= \frac{(3p)^2 + 2q(0)}{-q} = \frac{9p^2}{-q}$

$\frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} + \frac{\alpha\beta}{\gamma} = \gamma^2 + \alpha^2 + \beta^2 \quad \text{--- (1)}$
 $= (\gamma + \alpha + \beta)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = 0 - 2(3p) = -6p$

(11)

$$\begin{aligned} \textcircled{1} \quad &= (y + \alpha + \beta)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 0 - 2 \cdot 3p \\ &= -6p \end{aligned}$$

$$\frac{\beta\gamma}{\alpha} \cdot \frac{\alpha\gamma}{\beta} \cdot \frac{\alpha\beta}{\gamma} = \alpha\beta\gamma = -q \quad \text{--- } \textcircled{1}$$

$$\therefore \text{Required eqn is } x^3 + \frac{q}{p}x^2 - 6px + q = 0$$

$$\textcircled{ii} \quad \frac{\alpha\beta}{\gamma} = 1 \text{ is a root } \leftarrow \textcircled{1}$$

$$\therefore 1 + \frac{q}{p} - 6p + q = 0$$

$$\begin{aligned} q + \frac{q^2}{p} - 6pq + q^2 &= 0 \\ \therefore (3p - q)^2 + q &= 0 \quad \text{--- } \textcircled{1} \end{aligned}$$

$$\begin{aligned} \textcircled{c} \quad &\text{Let } P(x) = x^5 + 2x^2 + ax + b \\ &\text{If } (x+1)^2 \text{ is a factor } P(-1) = P'(-1) = 0 \\ &\therefore P(-1) = -1 + 2 - a + b = 0 \quad \leftarrow \textcircled{1} \\ &\therefore -a + b = -1 \\ &P'(x) = 5x^4 + 5x + a \quad \leftarrow \textcircled{1} \\ &P'(-1) = 5 - 4 + a = 0 \\ &a = -1 \\ &\therefore b = -2 \quad \leftarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} \textcircled{d} \quad &\sin(2k+1)\theta = \sin(2k\theta + \theta) \\ &= \sin 2k\theta \cos \theta + \cos 2k\theta \sin \theta \\ &\sin(2k-1)\theta = \sin(2k\theta - \theta) \\ &= \sin 2k\theta \cos \theta - \cos 2k\theta \sin \theta \\ &(\sin 2k\theta \cos \theta + \cos 2k\theta \sin \theta - \sin 2k\theta \cos \theta + \cos 2k\theta \sin \theta) \\ &\therefore \sin(2k+1)\theta = \sin(2k-1)\theta \\ &\text{or } \frac{\sin(2k+1)\theta - \sin(2k-1)\theta}{\sin \theta} = \frac{\sin \theta}{\sin \theta} \cdot \frac{2 \cos 2k\theta \sin \theta}{\sin \theta} \quad \left[\begin{array}{l} \sin(A+B) - \sin(A-B) = 2 \cos A \sin B \end{array} \right] \\ &= 2 \sin 2k\theta \cos(2k\theta) = 2 \cos(2k\theta) \sin \theta \quad \leftarrow \textcircled{1} \end{aligned}$$

:- Heron's Solution for Q5 (ii) (10b)

$$\frac{\beta\gamma}{\alpha} = \frac{\alpha\beta\gamma}{\alpha^2}$$

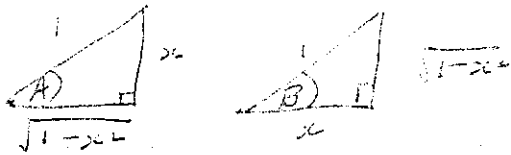
$$= \frac{q}{\alpha^2} \quad \text{Let } y = \frac{q}{\alpha^2} \quad \text{So } \alpha^2 = -\frac{q}{y} \quad \left. \begin{array}{l} \alpha = \sqrt{-\frac{q}{y}} \end{array} \right\} \text{--- } \textcircled{1}$$

$$\text{Sub } \textcircled{1} \rightarrow \text{eqn of polynomial: } x^3 + 3xp + q = 0$$

$$\begin{aligned} \text{i.e. } &\left(\sqrt{-\frac{q}{y}}\right)^3 + 3\left(\sqrt{-\frac{q}{y}}\right)p + q = 0 \\ &\sqrt{-\frac{q}{y}} \left[\left(\sqrt{-\frac{q}{y}}\right)^2 + 3p \right] + q = 0 \\ &\sqrt{-\frac{q}{y}} \left[\left(\sqrt{-\frac{q}{y}}\right)^2 + 3p \right] = -q \\ &-\frac{q}{y} \left[\frac{q}{y} + 3p \right] = -q \\ &-\frac{q}{y} \left[\frac{q^2}{y^2} - \frac{6pq}{y} + 9p^2 \right] = -q^2 \\ &-\frac{1}{y} \left[\frac{q^2}{y^2} - \frac{6pq}{y} + 9p^2 \right] = -q \\ &-\frac{q^2}{y^3} + \frac{6pq}{y^2} - \frac{9p^2}{y} = -q \\ &-q^2 + 6pqy - 9p^2y^2 = y^3q \\ &\text{So } qy^3 + 9p^2y^2 - 6pqy + q^2 = 0 \quad \text{Let } y = x \\ &\therefore x^3 + \frac{9p^2}{q}x^2 - 6px + q = 0 \end{aligned}$$

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(e) let $A = \sin^{-1}x$ and $B = \cos^{-1}x$
 $\therefore x = \sin A$ and $x = \cos B$



So $\sqrt{1-x^2} = \cos A$ $\sqrt{1-x^2} = \sin B$ ← ①
 $\therefore \text{LHS} = \sin(\sin^{-1}x - \cos^{-1}x)$
 $= \sin(A-B)$
 $= \sin A \cos B - \cos A \sin B$
 $= x \cdot \sqrt{1-x^2} - \sqrt{1-x^2} \cdot x$ ← ①
 $= x^2 - (1-x^2)$
 $= 2x^2 - 1$

QUESTION SIX :

(a)(i) $F = ma = 50 - kv^2$ ① limiting vel = 10
 $\therefore \ddot{x} = 0$
 when $v = 0$
 $k = \frac{1}{2}$

① So $2\ddot{x} = 100 - v^2$
 $\therefore 2 \frac{d^2x}{dt^2} = 100 - v^2$

(ii) $\ddot{x} = v \frac{dv}{dx}$
 $\therefore 2v \frac{dv}{dx} = 100 - v^2$ from part (c)

$\int \frac{2v}{100 - v^2} dv = \int \frac{dx}{x}$
 $\ln(100 - v^2) = -x + C$
 When $x = 0$
 $v = 0$
 $\therefore C = \ln 100$ ← ①
 $\therefore -x = \ln(100 - v^2) - \ln 100$

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$-x = \ln \left(\frac{100 - v^2}{100} \right)$
 $e^{-x} = \frac{100 - v^2}{100}$ ← ①
 $v^2 = 100(1 - e^{-x})$

(iii) $\ddot{x} = \frac{dv}{dt}$
 $\therefore 2 \frac{dv}{dt} = 100 - v^2$
 $\int \frac{dv}{100 - v^2} = \frac{1}{2} \int \frac{dt}{t}$ ← ①
 $\frac{1}{20} \int \left[\frac{1}{10 - v} + \frac{1}{10 + v} \right] dv = \frac{1}{2} \int \frac{dt}{t}$
 $\therefore -\ln(10 - v) + \ln(10 + v) = 10t + C$
 When $t = 0$
 $v = 0$
 $\therefore C = 0$

$t = \frac{1}{10} \ln \left(\frac{10 + v}{10 - v} \right)$ ← ①
 When $v = 5$ $t = \frac{1}{10} \ln \left(\frac{15}{5} \right) = \frac{1}{10} \ln 3$ seconds.

(iv) $\ddot{x} = -F + \frac{v^2}{2}$
 $v \frac{dv}{dx} = - \left(F + \frac{v^2}{2} \right)$
 $\int \frac{v}{(F + \frac{v^2}{2})} dv = - \int \frac{dx}{x}$ OR $\int \frac{2v}{2F + v^2} dv = \int \frac{2v}{2F + v^2} dv$
 $\ln \left(F + \frac{v^2}{2} \right) = -x + C$ ← ①
 When $v = 5$
 $x = 0$
 $\therefore C = \ln \left(F + \frac{25}{2} \right)$

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from part (ii) we obtain distance travelled when car reaches 5m/s.

$$25 = 100(1 - e^{-at})$$

$$\therefore e^{-at} = \frac{3}{4}$$

$$x = \ln\left(\frac{4}{3}\right)$$

Substitute $x = \ln\left(\frac{4}{3}\right)$ and $v = 0$ into eqn (1)

$$\ln\left(\frac{4}{3}\right) = \ln\left(\frac{F + \frac{25}{2}}{F}\right)$$

$$\therefore \frac{4}{3} = \frac{F + \frac{25}{2}}{F}$$

$$F = \frac{25}{2} = 37.5 \text{ N}$$

$$\frac{6}{4} = \frac{25 \text{ hr}}{4}$$

(b) Period $T = 2 \times [7:20 - 1:05] = 2 \times 375 = 750 \text{ min}$
 $a = 0.5(14.6 - 8.2) = 3.2 \text{ m}$
 since motion is SHM.

$$T = \frac{2\pi}{n} \text{ or } n = \frac{2\pi}{750} = \frac{\pi}{375}$$

$$x = 13.3$$

$$t = ?$$

Solⁿ of this eqn is $x = a \cos(nt + \phi)$ where $0 \leq \phi \leq 2\pi$

high 14.6m 7:20pm

$$\therefore x = 3.2 \cos(nt + \phi)$$

Considering the initial conditions

When $t = 0$ $x = -3.2$

$$\therefore -3.2 = 3.2 \cos(\phi)$$

$$\therefore \cos \phi = -1 \text{ or } \phi = \pi$$

$$x = 3.2 \cos(nt + \pi)$$

$$= -3.2 \cos nt$$

$$\text{Since } \cos(\phi + \pi) = -\cos \phi$$

When $x = -3.2$ $t = 0$

low 8.2m 1:05pm

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Min depth required = 13.3

$$x = 13.3 - 11.4 = 1.9$$

$$\therefore 1.9 = 3.2 \cos nt$$

$$\therefore nt = \cos^{-1}\left(\frac{1.9}{3.2}\right) = \pi - \cos^{-1}\left(\frac{1.9}{3.2}\right)$$

$$\text{or } t = \frac{1}{n} \left[\pi - \cos^{-1}\left(\frac{1.9}{3.2}\right) \right] = \frac{375}{\pi} \left[\pi - \cos^{-1}\left(\frac{1.9}{3.2}\right) \right]$$

$$\approx 263 \text{ min or } 4 \text{ hr } 23 \text{ min}$$

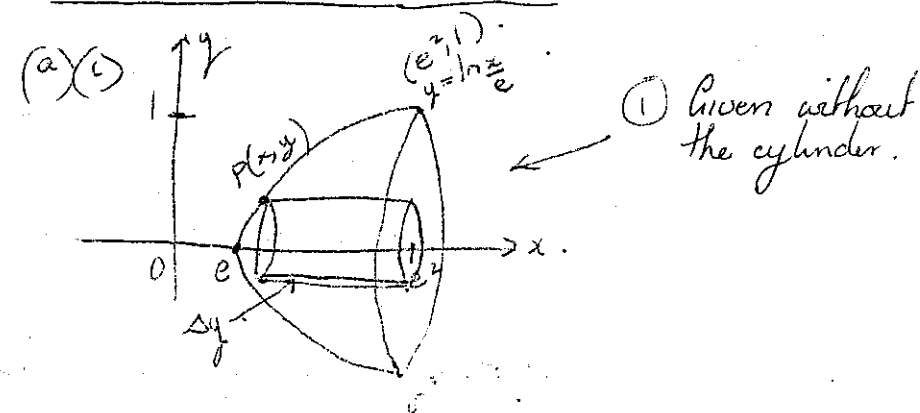
$$T - t = 750 - 263 = 487 \text{ min} = 8 \text{ hr } 7 \text{ min}$$

Hence the ship can leave the harbour

between 1:05 + 4:23 = 5:28 pm & }

1:05 + 8:07 = 9:12 pm

QUESTION SEVEN:



Given without the cylinder.

(ii) Considering hollow cylindrical shell:

radius y , height e^2

& thickness Δy

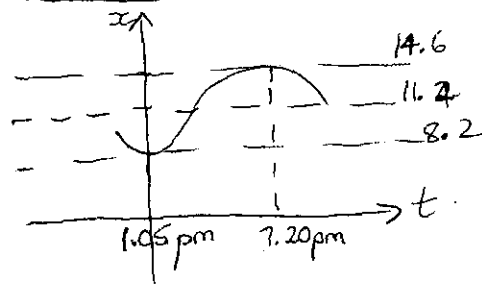
$$\therefore \Delta V = 2\pi y (e^2 - x) \Delta y$$

$$\text{Since } \ln \frac{x}{e} = y \text{ or } \Delta V = 2\pi y (e^2 - e^y)$$

$$\therefore \frac{x}{e} = e^y \therefore x = e^{y+1}$$

$$\text{Also for } x = e: u = 0 \text{ \& } \phi$$

Alternate Solution for Q4(b)



let x be the height of the tide at a particular time

Centre: $\frac{14.6 + 8.2}{2} = 11.4$
 Amplitude: $11.4 - 8.2 = 3.2$

Period: $375 \text{ mins} \times 2 = 750 \text{ mins}$ or $12\frac{1}{2} \text{ hrs}$ or $\frac{25}{4} \text{ hrs}$

then $\frac{2\pi}{n} = T$
 $\omega = \frac{2\pi}{n} = \frac{2\pi}{750}$
 $\frac{2\pi}{750} = n = \frac{\pi}{375}$
 or $n = \frac{4\pi}{25}$

- the height $x = 11.4 - 3.2 \cos \frac{4\pi}{25}t$
- the boat can leave when water height is greater than or equal to 13.3
 $13.3 \leq 11.4 - 3.2 \cos \frac{4\pi}{25}t$ (2nd/3rd Quad)

$\cos \frac{4\pi}{25}t = -\frac{19}{32}$
 then $\frac{4\pi}{25}t = 2.2065, 4.076$
 $\therefore t = 4.38, 8.11$

this converts to 4 hrs 22.8 mins, 8 hrs 11 mins
 Hence times the boat can leave the harbour:
 5:28 pm and 9:12 pm.

(16)

$V = 2\pi \int_0^1 y(e^y - e^{-y}) dy$
 $= 2\pi \int_0^1 y e^y dy - 2\pi \int_0^1 y e^{-y} dy$

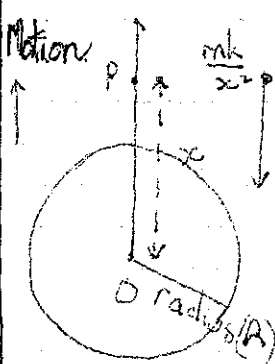
let $K = \int_0^1 y e^y dy$ use integration by parts
 $u = y, dv = e^y dy$
 $du = dy, v = e^y$

$K = [y e^y]_0^1 - \int_0^1 e^y dy$
 $= e - [e^y]_0^1 = 1$

So $J = \int_0^1 y dy = [\frac{y^2}{2}]_0^1 = \frac{1}{2}$

$V = 2\pi e^2 (\frac{1}{2}) - 2\pi e$
 $= \pi e^2 - 2\pi e$
 $= \pi e (e - 2)$

(b)(i)



Upward is pos. motion
 $m\ddot{x} = -\frac{mk}{x^2}$ or $\ddot{x} = -\frac{k}{x^2}$

On the surface of the earth
 $x = R$
 $-g = -\frac{k}{R^2}$, or $k = R^2 g$

Now $v \frac{dv}{dx} = -\frac{k}{x^2}$
 $\therefore \frac{dv}{dx} \left[\frac{1}{2} v^2 \right] = -\frac{R^2 g}{x^2}$

So $\frac{v^2}{2} = \frac{R^2 g}{x} + C$
 When $v = u$
 $x = R$

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then $\frac{1}{2}v^2 = \frac{R^2g}{x} + \frac{1}{2}u^2 - gR \leftarrow (1)$

$$v^2 = \frac{2R^2g}{x} + u^2 - 2gR$$

$$v^2 = u^2 + 2R^2g \left[\frac{1}{x} - \frac{1}{R} \right] \leftarrow (1)$$

(vel of a body at distance x from centre)

(ii) When the body is at the highest point $v=0$ and hence $0 = \frac{2R^2g}{x} + u^2 - 2gR$

$$\frac{2R^2g}{x} = 2gR - u^2$$

$$x = \frac{2R^2g}{2gR - u^2} \leftarrow (1)$$

$\therefore$ Greatest height H above the earth's surface

$$\text{is } H = \frac{2gR^2}{2gR - u^2} - R \leftarrow (1)$$

$$= \frac{2gR^2 - 2gR^2 + Ru^2}{2gR - u^2}$$

$$= \frac{Ru^2}{2gR - u^2}$$

(iii) Greatest height reached by the body will be infinite if $2gR - u^2 = 0$ or $u^2 = 2gR$

$$u = \sqrt{2gR}$$

Body will escape from the earth if

$$u > \sqrt{2gR} \leftarrow (1)$$

(iv) Min. escape velocity is given by $u = \sqrt{2gR}$

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$$u = \sqrt{2 \times 0.01 \times 6400} = 11.3 \text{ km/s correct to 1 dec place.}$$

(v) $u = \sqrt{2gR}$

$$v^2 = \frac{2R^2g}{x}$$

$$u v = \frac{\sqrt{2gR^2}}{\sqrt{x}}$$

$$u \frac{dv}{dt} = \frac{\sqrt{2gR^2}}{\sqrt{x}}$$

$$dt = \frac{\sqrt{2gR^2}}{\sqrt{2gR^2}} dx \leftarrow (1)$$

time for body to reach a height of $3R$ above the earth (then $x = 4R$) from initial time $t=0$ (when $x=R$) is given by

$$(1) \quad t = \int_R^{4R} \frac{x^{1/2}}{\sqrt{2gR^2}} dx$$

$$= \frac{1}{\sqrt{2gR^2}} \left[\frac{2}{3} x^{3/2} \right]_R^{4R} \leftarrow (1)$$

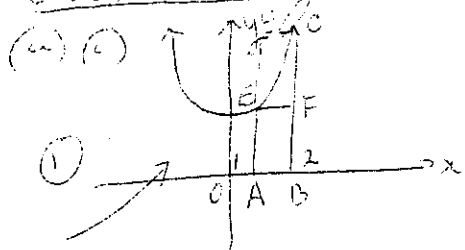
$$= \frac{1}{\sqrt{2gR^2}} \left[\frac{2}{3} \times 4^{3/2} R^{3/2} - \frac{2}{3} R^{3/2} \right]$$

$$= \frac{2}{3} \times \frac{1}{\sqrt{2gR^2}} \times 7R^{3/2}$$

$$= \frac{14}{3} \sqrt{\frac{R}{2g}}$$

Question Eight.

(19)



(a) (i) Area of ABFE $< \int (px^2 + q) < \text{Area of } ABC$

(ii) $\frac{AB \times AE}{1 \times (p \times q)} < \left[\frac{px^3}{3} + qx \right]_1^2 < AB \times BC$

(i) $\frac{AB \times AE}{1 \times (p \times q)} < \left[\frac{8p}{3} + 2q \right] - \left(\frac{p}{3} + q \right) < 1 \times (4p)$

$\therefore (p+q) < \frac{7p+3q}{3} < 4p$ — (1)

(b) (c) In HPRM, $\angle HPR = \angle RMH = 90^\circ$
 (1) $\rightarrow$ (Angle between radius & tangent at pt. of contact $= 90^\circ$)

(i) $\left\{ \begin{array}{l} \therefore \text{HPRM is a cyclic quad. (opp } \angle \text{ s are suppl.)} \\ \text{Similarly for MRQK.} \end{array} \right.$

(ii) Construction: Join PM and QM on

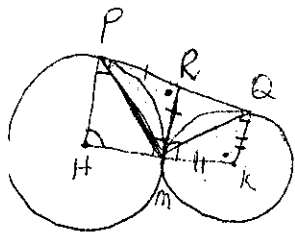
(1) $\rightarrow \left\{ \begin{array}{l} \triangle PRM \text{ \& } \triangle MQK \\ \angle PRM = \angle MKQ \\ \text{(Ext. } \angle \text{ of cyclic quad. MRQK)} \end{array} \right.$

$PR = RM$ (tangent from ext. pt.)
 $KM = KQ$ (radii)
 $\therefore$ triangle is isosceles.

$\therefore \angle RPM = \angle RMP = \angle KMQ$

$\therefore \triangle PRM \parallel \triangle MQK$

(1) $\rightarrow$ (equiangular)



(c) (i) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ — (20)

$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

$e^{ix} = 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \dots$

(1) $e^{ix} = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) + i \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$
 $= \cos x + i \sin x$

$e^{i\pi} = \cos \pi + i \sin \pi = -1$

(1) $e^{i\pi} = -1$ since $\cos \pi = -1$ & $\sin \pi = 0$

(ii) $e^{\frac{i\pi}{2}} = (e^{i\pi})^{\frac{1}{2}} = (-1)^{\frac{1}{2}} = i$ — (1)

from (i) $i = [e^{\frac{i\pi}{2}}]^i = e^{-\frac{\pi}{2}}$ which is real. — (1)

(iii) $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$
 $\ln z = \ln r + i\theta$
 $= \ln r + i\theta$ — (1)

This will give an infinite number of results unless θ is restricted.

$\theta = \ln r + i(\theta + 2\pi k)$ where k is any integer.

The principal value of $\ln z$ occurs when $-\pi \leq \theta \leq \pi$. — (1)

(iv) $z = 1+i$
 $\ln z = \ln \sqrt{2} + \frac{i\pi}{4}$ — (1)